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Aakanksha Melkani

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EDUCATION

Ph.D., Agricultural Economics, Michigan State University, USA *2016-2021*

M.Phil., Development Studies, Indira Gandhi Institute of Development Research, India *2014-2016*

M.Sc., Agricultural Economics, Punjab Agricultural University, India *2012-2014*

B.Sc., Agricultural Sciences, G.B. Pant University of Agriculture and Technology, India *2008-2012*

EMPLOYMENT

Assistant Professor of Economics *September 2024 to present*
Siena College

Post Doctoral Research Associate *Oct 2021 to August 2024*
University of Nebraska

Graduate Research/Teaching Assistant *Aug 2016 to Aug 2021*
Michigan State University

Other relevant professional experience: Quality controller and master trainer for national Rural Agricultural Livelihoods Survey (RALS) household panel data collection in Zambia *May-July, 2019*

RESEARCH

Peer-reviewed publications

- Melkani, A., Mason, N. M., Mather, D. L., Chisanga, B., and Jayne, T. (2025). Liquidity constraints for variable inputs at planting time and the maize production and marketing decisions of smallholder farmers in Zambia. *Agricultural Economics*, 56(1), 73-91. Available here
- Melkani, A., and Kataria, P. (2015). The development-conservation trade-off facing Indian agriculture: Carbon costs of Punjab agriculture. *Agricultural Economics Research Review*, 28, 219-228. Available here
- Melkani, A. (2014). Intensive agricultural practices of rice-wheat system in Punjab: Implications on costs and productivity. *Indian Journal of Economics and Development*, 10(2), 132-139. Available here

Work in progress

- Economic impact of groundwater regulations in Nebraska: A hedonic price analysis (with Taro Mieno, Aaron Hrozencik, Renata Rimsaite, Shunkei Kakimoto and Nicholas Brozovic) | first author. Working paper available here
- The Impact of Drought on Farmland Values through a Hedonic Price Analysis of Farmland Transactions in Contiguous US (with Taro Mieno, Aaron Hrozencik, Renata Rimsaite, and Nicholas Brozovic) | first author. Working paper available here
- Drought- and heat-related crop insurance outcomes in the presence of groundwater regulations: Evidence from Nebraska (With Sungmin Cheu)
- How do water laws shape the adaptation to drought and water scarcity? The case of crop land use across the Republican River Basin (with Tallulah Powers)

OUTREACH and ENGAGEMENT

Select Academic Paper Presentations

- Drought- and heat-related crop insurance outcomes in the presence of groundwater regulations: Evidence from Nebraska: Association of Environmental and Resource Economics Annual Meet 2025.
- The Impact of Drought on Farmland Values through a Hedonic Price Analysis of Farmland Transactions in Contiguous US: Association of Environmental and Resource Economics Annual Meet 2024.
- Economic impact of groundwater regulations in Nebraska: A Hedonic price analysis: Agricultural and Applied Economics Association (AAEA) Annual Meeting 2023; Southern Economic Association (SEA) Annual meeting 2023; Seminar in Water Economics OnLLine (SWELL) 2023.
- Smallholder maize market participation and choice of marketing channel in the presence of liquidity constraints: Evidence from Zambian maize markets: Midwest International Economic Development Conference (MWIEDC) 2021 (virtually); International Conference of Agricultural Economists (ICAE) 2021 (virtually)

Poster presentations: AAEA annual meeting 2019 and 2020; Heartland Environmental and Resource Economics Workshop 2022

Invited talks: "Sustainability in the Environmental Economics classroom", at the Siena College 2025 Sustainable Classrooms Workshop, May 21-22 2025.

Non-academic presentations at events organized by collaborating organizations:
- Indaba Agricultural Policy Research Institute, Lusaka, Zambia (August, 2020)
- USDA, Economic Research Service (April, 2023)

Published outreach products based on ongoing or past research projects
- Blog: Economic Impact of Groundwater Regulation: Case Study of Nebraska. Available here
- Policy brief: Smallholder maize market participation and choice of marketing channel in the presence of liquidity constraints: Evidence from Zambia. Available here
- Blog: What do we know about groundwater access and farmland values. Available here
- Info-graphics on current irrigation patterns in the USA. Available here and here
- Podcast: Invited podcast with the Middle Republican Natural Resources District on my research and field work, here
- Media coverage: Interviewed by Yanqi Xu of the *Flatwater Free Press*. She highlighted my work on the valuation of groundwater and the impact of groundwater regulations on farmland values in Nebraska in the Dec. 21, 2023 article "Water's worth: It sits beneath Nebraska's farmland and has serious value. But who owns it?"

Others - Presented simplified version of PhD research for engagement with prospective incoming graduates at Michigan State University (April, 2021)

TEACHING

Siena University

- **Principles of Microeconomics, Econ 101** Fall 2024, Spring 2025; 3 credits; 25-33 students/semester:
  This is a required core course for all business majors at Siena College. It includes introductory topics such as demand, supply, price determination, market structures, producer and consumer theory with an emphasis on application to current affairs. I developed new course material for the class in Fall 2024. I continually update the course material, particularly the class assignments to better connect class concepts with current economic issues.
- **Environmental Economics, Econ 340/Enva 300** Spring 2025; 3 credits; 18 students:
  This is a required course for all Environmental Studies students at Siena College and an elective for Economics majors. I developed the course in three modules: (i) economic theory (externalities, Coase theorem, property

rights, public goods); (ii) environmental policy (decentralized, command and control, incentive based, and market based policies); (iii) empirical benefit-cost analysis of environmental policy (hedonic price analysis, averting behavior, stated preference methods, etc.).

- **Applied Economic Analysis, Econ 300** Fall 2025; 3 credits; 14 students:
  This course was designed and offered for the first time in the Department of Economics, as a pilot to a future intermediate level course in economic methods. Key topics covered included (i) Math for Econ including linear algebra and derivatives and their applications to Econ problems (ii) Descriptive statistics (iii) Inferential statistics.

University of Nebraska-Lincoln

- **R lab instructor** for the Applied Econometrics course at the University of Nebraska-Lincoln (Fall 2022). Responsibilities included primary instruction and partial content development for weekly R labs.

Michigan State University

- **Data Analysis for Agri-Food System, AMB 203** Spring 2021; 75 students; Responsible for primary instruction, course management, and partial content development.
- Teaching assistant and grader for various undergraduate courses across multiple semesters. Graded homework, quizzes and exams, and held weekly office hours. courses

HONORS AND AWARDS

Received an **'Honorable mention'** for the **best presentation/paper** during the **Graduate Research Symposium 2020** held by the Department of Agricultural, Food, and Resource Economics, MSU, for the paper, "Smallholder maize market participation and choice of marketing channel in the presence of liquidity constraints: Evidence from Zambia"

Winner of the **R.T. Doshi Foundation Award for Best Presentation** during the **23rd Annual Conference of Agricultural Economics Research Association** (Mumbai, 2016) for the paper, "The development-conservation trade-off facing Indian agriculture: Carbon cost of Punjab agriculture"

Recipient of the **National Junior Research Fellowship** for meritorious students from the **Indian Council of Agricultural Research (ICAR-JRF)** for the academic years 2012-2014

SERVICE

Faculty track judge for the Econometrics track session at the 19th Annual Ted Winnowski '63 Student Conference in Business 2025, Siena College (May 2, 2025)

Judged undergraduate and graduate poster competitions at Research Days, University of Nebraska-Lincoln, 2022 and 2023

Vice President - Graduate Student Organization and member of the **Graduate Policy Committee** at the Department of Agricultural, Food, and Resource Economics, Michigan State University (May, 2020 to May, 2021)

Refereeing for Journal Articles - Journal of African Economies (2023); Cogent Food and Agriculture (2024); Agricultural Economics (2024); Journal of Development Perspectives (2025); Frontiers in Sustainable Food Systems (2026).

Refereeing for conference submissions - AAEA annual meet, 2023,2024,2025,2026; AERE annual meet, 2023 and 2024; IAAE annual meet, 2020.

Professional membership - Agricultural and Applied Economics Association (AAEA); Western Agricultural Economics Association (WAEA); International Association of Agricultural Economists (IAAE); Association of Environmental and Resource Economists (AERE)

GRANTS and FELLOWSHIPS

Summer Scholars 2025 one-time grant of $1000 for supporting research activity with an undergraduate student during Summer 2025; Funded through the Center for Undergraduate Research and Creative Activity (CURCA), Siena College.

Siena College's pedagogy fellowship, 2025 for the development of a new economic methods course to help students translate core mathematical and statistical competencies to intermediate and upper-level economics courses. Funding amount:$2500; Funded through the Committee on Teaching and Faculty Development (COTFD), Siena College for Summer 2025.

Wrote the concept note and developed associated budget for the **lab-in-the-field** component of several grant proposals submitted to the Bill and Melinda Gates Foundation, OCP Foundation, USAID, and USDA Foreign Agricultural Service (Fall 2019).

ADDITIONAL SKILLS

Technical skills: Proficient in STATA, LaTeX, and R; Version control using GitHub; Working knowledge of MATLAB, SPSS, and GAMS
Languages: Hindi (native); English (fluent); Spanish (beginner); working knowledge of Punjabi and Bengali